

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 25: Probability

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Probability of an event

1. Which of these cannot be the probability of an event? (a) $\frac{5}{4}$ (b) 0.25 (c) $\frac{1}{33}$ (d) 67%

ICSE 2025, 1 mark

SOLUTION

A probability always lies from 0 to 1. $\frac{5}{4} = 1.25$ is greater than 1.

Answer: (a)

2 Balls, numbered cards and letters

2. A box holds cards numbered 1 to 30, and one card is drawn at random. A wall clock is won for a perfect square, a water bottle for an even number that is also a multiple of 3, and a purse for a prime number. Find the probability of winning (a) the wall clock (b) the water bottle (c) the purse.

ICSE 2025, 3 marks

SOLUTION

There are 30 equally likely outcomes.

(a) Perfect squares: 1, 4, 9, 16, 25 (5 cards). Probability = $\frac{5}{30} = \frac{1}{6}$.

(b) Even multiples of 3 are the multiples of 6: 6, 12, 18, 24, 30 (5 cards). Probability = $\frac{5}{30} = \frac{1}{6}$.

(c) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (10 cards). Probability = $\frac{10}{30} = \frac{1}{3}$.

Answer: (a) $\frac{1}{6}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$

3. The letters A, D, M, N, O, S, U, Y are written on separate cards. One card is drawn at random. Find the probability that the letter (a) is in the word MONDAY (b) is not in MONDAY (c) is in both SUNDAY and MONDAY. **ICSE 2024, 3 marks**

SOLUTION

There are 8 cards.

(a) Letters of MONDAY: M, O, N, D, A, Y, which is 6 cards. Probability = $\frac{6}{8} = \frac{3}{4}$.

(b) $1 - \frac{3}{4} = \frac{1}{4}$ (the letters S and U).

(c) Common to both words: N, D, A, Y, which is 4 cards. Probability = $\frac{4}{8} = \frac{1}{2}$.

Answer: (a) $\frac{3}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$

4. A letter is chosen at random from the English alphabet. Find the probability that it is a vowel. (a) $\frac{4}{26}$ (b) $\frac{5}{26}$ (c) $\frac{21}{26}$ (d) $\frac{5}{24}$ **ICSE 2023, 1 mark**

SOLUTION

There are 26 letters and 5 vowels (a, e, i, o, u).

Answer: (b)

5. A bag holds 25 cards numbered 1 to 25, and one card is drawn at random. Find the probability that its number is (a) a multiple of 5 (b) a perfect square (c) a prime number. **ICSE 2023, 3 marks**

SOLUTION

(a) Multiples of 5: 5, 10, 15, 20, 25 (5 cards). Probability = $\frac{5}{25} = \frac{1}{5}$.

(b) Perfect squares: 1, 4, 9, 16, 25 (5 cards). Probability = $\frac{1}{5}$.

(c) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23 (9 cards). Probability = $\frac{9}{25}$.

Answer: (a) $\frac{1}{5}$ (b) $\frac{1}{5}$ (c) $\frac{9}{25}$

6. Each letter of the word AUTHORIZES is written on a separate disc, and one disc is drawn at random. Find the probability that the letter is (a) a vowel (b) one of the first 9 letters of the alphabet (c) one of the last 9 letters of the alphabet. **ICSE 2020, 3 marks**

SOLUTION

The word has 10 different letters.

(a) Vowels: A, U, O, I, E (5 discs). Probability = $\frac{5}{10} = \frac{1}{2}$.

(b) The first 9 letters are A to I. In the word: A, H, I, E (4 discs). Probability = $\frac{4}{10} = \frac{2}{5}$.

(c) The last 9 letters are R to Z. In the word: U, T, R, Z, S (5 discs). Probability = $\frac{5}{10} = \frac{1}{2}$.

Answer: (a) $\frac{1}{2}$ (b) $\frac{2}{5}$ (c) $\frac{1}{2}$

3 Coins and dice

7. Assertion (A): When a die is thrown once, the probability of getting an even number is $\frac{2}{3}$. Reason (R): The even numbers on a die are {2, 4, 6}. (a) A true, R false (b) A false, R true (c) both true (d) both false **ICSE 2024, 1 mark**

SOLUTION

There are 3 even numbers out of 6, so the probability is $\frac{3}{6} = \frac{1}{2}$. A is false.

R is true.

Answer: (b)

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.